PS 3: Week 9

From Samples to Populations

Maya Mikelson

Agenda

1. Generalizability and Weighting
2. Descriptive Hypothesis Testing

From Samples to Populations

- **Experimental**
  - We run an experiment exposing people to campaign materials and measure the effect of these materials on someone's likelihood of voting for our candidate
    - How do we know this generalizes across cases?
- **Descriptive**
  - We run a poll measuring support for our candidate
    - How do we know this generalizes across our population?

Weighting

Category	Weight	Student's Grade	Weighted Grade
In-class Individual Assignments	0.10	75%	7.50%
In-class Group Assignments	0.05	90%	4.50%
Problem Sets	0.30	95%	28.50%
Final Project	0.10	90%	9.00%
Peer Evaluations	0.05	100%	5.00%
Section Attendance	0.05	100%	5.00%
Midterm	0.15	90%	13.50%
Final	0.20	90%	18.00%
Total			**91.00%**

Weighted grade = weight $\times$ grade, summed across categories.

Attendance Survey



Generalizing From Populations

- We take samples from a population because we want to learn something about that population
- But samples are always prone to some kind of sampling bias!
  - **Sample selection bias:** who do we ask to take the survey?
  - **Non-response bias:** what kind of people take (or don't take) political surveys?
- And what about random chance?
  - E.g. a sample I take on a Monday will be different from a sample I take on a Tuesday
- You already have the tools to address these problems, what are they?

Generalizing From Populations

- We take samples from a population because we want to learn something about that population
- But samples are always prone to some kind of sampling bias!
  - **Sample selection bias:** who do we ask to take the survey?
  - **Non-response bias:** what kind of people take (or don't take) political surveys?
- And what about random chance?
  - E.g. a sample I take on a Monday will be different from a sample I take on a Tuesday
- You already have the tools to address these problems, what are they?
  - **Sampling bias:** weighting, probability sampling
  - **Random chance:** 95% confidence intervals, increasing sample size, standard error, p-values

Descriptive Hypothesis Testing: Is this Random Chance?

- We collected some data and found that Group A is different from Group B.
  - Ex.: Younger voters support our candidate more than older voters.
- There is still the same issue as before:
  - Is the observed difference between Group A and Group B due to random chance (from sampling), or is it because the two groups truly are not the same?
- Note: Our goal with this approach is to describe differences, **NOT** determine causality! We would need to do an experiment to prove that (remember Heterogeneous Treatment Effects)

Descriptive Hypothesis Testing w/ Weights: Addressing Sampling Bias

- Some people are more likely to answer our surveys/partake in our experiments than others
  - We can weight groups that are underrepresented in our sample *more* to provide better balance in our data
  - We can weight groups that are overrepresented in our sample *less* to provide better balance in our data
  - Ex: if college-educated voters are overrepresented in our sample, they get a smaller weight; if non-college-educated voters are underrepresented in our sample, they get a bigger weight
- Adding weights to our descriptive hypothesis tests can help us take into account sampling bias when comparing two groups

Acknowledgements

Special thanks to prior GSIs for providing materials off of which some of these slides are based, especially Will Halm.